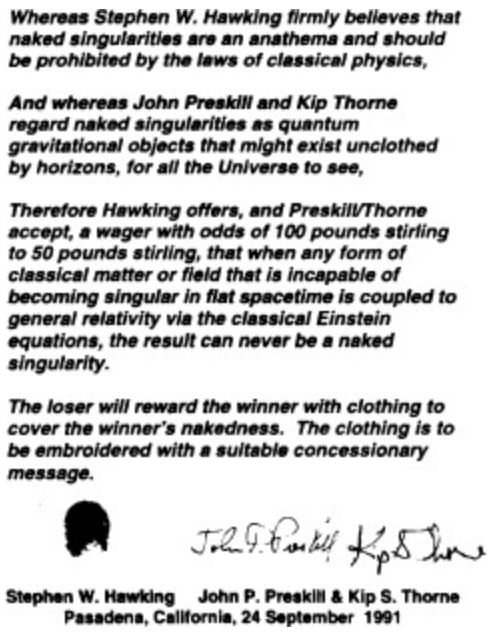
forming in the first place. So it seemed. However, neither Penrose's mathematical manipulations nor anybody else's were powerful enough to say for sure.

In 1969 Penrose, strongly convinced that naked singularities cannot form, but unable to prove it, proposed a conjecture, *the conjecture of cosmic censorship: No imploding object can ever form a naked singularity; if a singularity is formed, it must be clothed in a horizon so that we in the external Universe cannot see it*

Members of the physics "establishment"—physicists like John Wheeler, whose viewpoints are the most influential—have embraced cosmic censorship and espouse it as almost surely correct. Nevertheless, nearly a quarter century after Penrose proposed it, cosmic censorship remains unproved; and recent computer simulations of the implosion of highly nonspherical stars suggest that it *might* even be wrong. Some implosions, according to these simulations by Stuart Shapiro and Saul Teukolsky of Cornell University, might actually create naked singularities. Might. Not will; just might.

Stephen Hawking is the epitome of the establishment these days, and John Preskill (a colleague of mine at Caltech) and I enjoy tweaking the establishment a bit. Therefore, in 1991 Preskill and I made a bet with Hawking (Figure 13.8). We bet that cosmic censorship is wrong; naked singularities *can* form in our Universe. Hawking bet that cosmic censorship is right; naked singularities can never form.



Whereas Stephen W. Hawking firmly believes that naked singularities are an anathema and should be prohibited by the laws of classical physics,

And whereas John Preskill and Kip Thorne regard naked singularities as quantum gravitational objects that might exist unclothed by horizons, for all the Universe to see,

Therefore Hawking offers, and Preskill/Thorne accept, a wager with odds of 100 pounds stirling to 50 pounds stirling, that when any form of classical matter or field that is incapable of becoming singular in flat spacetime is coupled to general relativity via the classical Einstein equations, the result can never be a naked singularity.

The loser will reward the winner with clothing to cover the winner's nakedness. The clothing is to be embroidered with a suitable concessionary message.

Stephen W. Hawking John P. Preskill & Kip S. Thorne
Pasadena, California, 24 September 1991